Personal, Background, and Future Goals Statement

On the back of my door is a poster titled “map of mathematics”, a vibrant diagram with subjects from logic to vector calculus to statistics. I am fascinated by math. In my opinion, math teaches us invaluable lessons, including how to create well-defined problem statements, but also how to think profoundly and creatively. This reasoning motivated my course of study at the University of Rochester, and in May 2023, I graduated with an Honors Bachelor of Science in mathematics.

To the left of my bed is a poster of Half Dome in Yosemite National Park. The summer after graduating college, I was a backpacking guide in Yosemite. Through this experience, I became intimately familiar with changes in the park's environment. I learned about dwindling snowpack, rising temperatures, vegetation shifts, and wildlife patterns. Monitoring and predicting these changes are crucial, both for leading a successful backpacking trip and, on a large scale, for understanding how climate change threatens the park.

It took me until the last year of my undergraduate studies to reconcile these two fascinations. At second glance, in the upper right-hand corner of my “map of mathematics” poster, there is a small section that says “Computer Science”, and under that, “Machine Learning”. Machine learning (ML) models, specifically geospatial machine learning (GeoML) models, are increasingly used for environmental monitoring and prediction. **I want to develop new algorithms and frameworks to improve GeoML models, specifically those used for environmental monitoring and prediction.**

Intellectual Merit

I value the profound understanding I have of math, which is reflected in my success in coursework and my mathematical research. To recognize my proficiency in mathematics and my promise in the field, **my undergraduate institution presented me with awards during my sophomore, junior, and senior year**: **the Stoddard Prize**, awarded to two students annually, the **Doris Ermine Smith Award**, awarded to one student annually, and lastly, the **Arthur S. Gale Memorial Prize**, awarded to one student receiving a Bachelor's degree in mathematics that, in the judgement of the Department, displays the greatest promise in the field. These awards made me confident in my mathematical abilities and motivated me to consider how I want to use my theoretical knowledge. By my final research project during my undergraduate career, I had established my research interests in statistical machine learning. I look to continue my research in statistical machine learning, with the goal of developing and improving algorithms and frameworks for geospatial problems. My research in machine learning is driven by the toolbox I built during my previous math research experiences.

Undergraduate Research: The summer after my sophomore year, I participated in my first REU: the Coding Theory REU at Occidental College. Our research involved the comparison of theta series associated to codes and lattices in real quadratic fields, and we proved a new bound on the similarity of theta series associated to different lattices. This result can be used to study the interplay between the coding theory, lattice theory, and number theory, with applications to theoretical computer science. For this experience, my peers and I created a novel program to compare theta series associated to different lattices, which demonstrated the utility of computer programs, even in math research. Following this REU, we presented our research at the Southern California Math REU Conference. My project group continued working on our research throughout the following year, and our paper was **accepted for publication in the journal *Designs, Codes, and Cryptography***.

While I enjoyed this research experience, I recognized that I wanted my future research to have more direct applications to the real world. The following year, I applied and was accepted to the Combinatorics, Geometry, and Data Science group in the SMALL REU at Williams College. During this time, I was able to work on three different research projects, and I was most passionate about my final project due to its relevance to real-world problems.

One of my projects investigated a problem involving Benford's Law. For this problem, a d-dimensional box is fragmented by drawing proportions from a random process. We conjectured that as the number of iterations of this process increased, lower-dimensional measurements of the resulting hyperboxes tended toward a Benford distribution. I took interest in this project as Benford's law fascinates mathematicians and non-mathematicians alike, appearing in datasets involving taxes, house prices, genome data, and academic publishing networks. For this project, I developed code simulating this process to examine the distribution of these resulting measurements, which, again, illustrated the usefulness of computer programs in my research. By the end of the summer, I learned an extensive amount of probability theory, which we used to prove this conjecture. Following this research experience, **my colleague and I presented this work at the Young Mathematician's Conference at Ohio State University**. In October of 2022, **we presented this work at the Advances in Interdisciplinary Statistics and Combinatorics conference at UNC Greensboro**, where **we were awarded Best Undergraduate Presentation**. Following this conference, in the spring of 2023, **we submitted a manuscript to the Journal of Statistical Theory and Practice's special issue**.

Another of my projects examined a variation of the Erdős problem involving angles in 3D space. We looked to answer: what is the minimum number of distinct angles created by a set of n points in 3D space? In this project, we used the restriction of general position, a condition that prohibits three points on a line and four points on a circle. Obeying this restriction, we found different constructions of points in 3-dimensions that allowed us to determine an upper bound on the minimum number of distinct angles, and we developed a new python program that took a configuration of points and determined the number of distinct distances. To determine the lower bound, we were able to relate this research to the problem of determining the number of distinct distances on a sphere. This research focused primarily on geometric configurations, which was a novel mathematical pursuit for me. Following this research, **our paper "Distinct Angles and Angle Chains in Three Dimensions" was accepted for publication in the journal Discrete Mathematics and Theoretical Computer Science.**

While these experiences strengthened my love for research and provided me with valuable skills, I was most passionate about my third project due to its applications beyond math—specifically to machine learning problems. In this project, we studied VC-dimension, a measure of complexity which gives way to the fundamental theorem of statistical learning and determines the Probably-Approximately-Correct (PAC) learnability of a set of functions; PAC-learnability forms the foundation for designing and analyzing various machine learning algorithms. In our project, we investigated the VC-dimension of spherical hypothesis classes, which yielded an interesting complexity problem from the point of view of learning theory. Throughout this project, I read *Understanding Machine Learning: From Theory to Algorithms* by Shai Shalev-Shwartz and Shai Ben-David. From this book, I learned about algorithms and learning models through a statistical framework, which gave me a new perspective for my mathematical research. Following this REU, **our paper "VC-dimension and Distance Chains in F_q^d" was accepted for publication in the *Korean Journal of Mathematics***. During my senior year, I further generalized the theory in this paper to hypothesis classes with an arbitrary number of parameters and **presented this work in my Honors Thesis *VC-dimension of Spherical Hypothesis Classes over F_q^d*.**

<u>**Deciding how to use my mathematical knowledge**</u>: I am deeply indebted to these research opportunities for encouraging my mathematical maturity and allowing me to hone my research interests. Following my last project, I was inspired to continue my research in statistical machine learning, but I maintained the goal of seeing my work applied to solve environmental problems. For these reasons, I delayed my applications to graduate school and took time to search for a field where I could pursue this goal; I read literature from different fields of study, until I found geospatial machine learning, a field that uses statistical machine learning methodology for environmental monitoring and prediction tasks. With these interests in mind, **I applied and was accepted to the University of Colorado at Boulder's PhD program in computer science**, a prime location for studying geospatial machine learning due to its top tier research program and opportunity to collaborate with scientists from related domains.

<u>**Graduate Education:**</u> I am advised by Esther Rolf, who has performed innovative research that applies methodological techniques from statistical machine learning to geospatial machine learning problems. Specifically, I am working on developing a statistical sampling framework to automate the process of generating high-quality training sets for geospatial machine learning models. **My work is motivated by the goal of improving the performance of geospatial machine learning models that are applied to**

social and environmental monitoring and prediction tasks. Since being accepted to the PhD program, the University of Colorado at Boulder has awarded me with the **Chancellor's Fellowship**, a prestigious fellowship awarded to outstanding incoming students, which will provide partial funding for the first and last years of my graduate studies.

Broader Impacts

As an undergraduate researcher, I frequently looked to my mentors in both math and computer science for advice. The lack of representation for women in these fields was discouraging. As an undergraduate, I only had one female math professor. None of my computer science professors were women. Even among my peers, I was often reminded that I was a woman in STEM. In my graph theory class, I was the only student who was a woman. On one of my research projects, I was the only woman out of 6 other students and 3 advisors. The disproportion between women and men researchers in both math and computer science is apparent in my own experiences. Studies have shown that having a female mentor has a positive effect on the performance of female undergraduates, yet in these fields, there are very few female faculty members that can serve as such mentors[1]. In my graduate career**, I plan to work with the Association for Women in Science, the CS Graduate Student Association, and the department's DEI Committee** to organize events for women and underrepresented groups in the CS PhD program.

During my graduate education, I also plan on seeing broader impacts through teaching. Many of my proudest moments during my undergraduate career came from being a teaching assistant for a freshman calculus class. For this position, I led workshops during which students could get help on their homework. Additionally, I frequently had students ask me for advice on course schedules, declaring majors, and how to get involved in research. In this way, being a teaching assistant for this course also doubled as being a peer-advisor for these students. This position made me recognize the significance of positive experiences in STEM during a student’s academic career. After I graduated, I worked as a teacher at an after-school learning center. This gave me an opportunity to create positive experiences in STEM for younger students. Many of the students that came into this center lacked confidence in school, and I worked to provide these students with personal attention and instill confidence in their work. Whether it’s teaching undergraduates or middle and high school students, teaching is an important part of my academic career that connects my skill set and interests with the needs of my broader community. **I will continue creating positive experiences in STEM by volunteering for the CU Science Discovery program as a STEM mentor**, and in this program, I will mentor elementary school students in math and STEM.

Future Goals

My proposed research will create effective and feasible data sampling strategies for ML-informed data collection, which will improve performance of GeoML models. **This work is inherently interdisciplinary**. The NSF-GRFP will grant me the freedom to conduct this work and allow me to collaborate with domain scientists in and outside of the computer science community. After my graduate education, I will pursue a career in academia, where I will continue using cutting-edge machine learning research to contribute to current environmental monitoring and prediction efforts. As a female faculty member in computer science, I plan to mentor women and under-represented groups in STEM through undergraduate and graduate research. I intend to lead my own lab, and, through the opportunities in this position, I will continue mentoring the next generation of mathematicians and computer scientists.

References

[1] National Mathematics Survey (2019) MIT